

TRUNKBELAY · REQUIREMENTS

TrunkBelay requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/trunkbelay		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	Amish's decisions 16A, 17B, 18C and 19A carried out (TKB-DDR-003); R9 target restated by Amish; status of R2, R4, R7 and R9 updated
0.2	2026-10-03	Amish Chadha	Status of every requirement on paper at TRL 3 (TKB-CAL-001); targets unchanged
0.1	2026-09-30	Amish Chadha	Initial scaffold

TrunkBelay requirements

Requirements with their status on paper at TRL 3. The targets are unchanged from version 0.1 except R9, which Amish restated on 2026-10-03; they are to be confirmed with the co-design partner. The status column comes from the calculation note TKB-CAL-001 v0.2.

Amish Chadha, 2026-10-03, on the recommendations put to him: "i agree with all the 46 recommendations you provided. please proceed." For TrunkBelay that is decisions 16A (R2: keep the drill rule, double-acting collar as the recorded fallback), 17B (R4: pack the victim set pre-rigged), 18C (R7: the helpers haul the rope bag on a tag line and the collar frame goes to aluminium) and 19A (R9: keep the auto-locking descender and evacuation triangle, seek group purchase, restate the cost target). The design decisions register (TKB-DEC-001) and decision record TKB-DDR-003 hold the detail.

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status on paper (TRL 3)
R1	Collar holds on a palm trunk	No slip greater than 20 mm under a 2.5 kN (562 lbf) static downward load on wet trunk sections	Proof test on CalRig with cut trunk sections at TRL 3	Met on paper; the friction on wet bark is to be confirmed (locking factor 1.54, settles about 10 mm)
R2	Collar holds when load direction changes	No slip greater than 50 mm when a 100 kg (220 lb) test mass passes the collar during lowering	Drop and transition tests on trunk sections; the upward case as a misuse test only	At risk for an upward pull: the single top chain cannot hold it. Decided by Amish (16A): the drill rule stays (collar at least 300 mm above the victim's attachment); the double-acting collar is the recorded fallback if the TRL 4 drill shows a rescuer cannot reach above the attachment
R3	Fits common palms	Trunk diameters 200 to 450 mm (8 to 18 in) without tools	Fit trials on surveyed palms	Met: the V and the 110-link chain fit the whole range
R4	Quick to rig	Collar fixed and victim clipped in under 5 minutes by a trained climber	Timed drills at 3 m height	At risk: about 4.5 minutes estimated, a thin margin; the pre-rigged victim set (17B) saves 45 s and the tag-line haul (18C) adds 45 s
R5	Controlled lowering	Descent speed held under 0.5 m/s with hand force under 150 N (34 lbf) for a 100 kg (220 lb) load	Lowering tests with test mass	Met on paper: about 80 N at the hand with the brake carabiner
R6	Rope length	Lowers from 25 m (82 ft)	Measured	Met: 30 m of rope, 28 m needed
R7	Portable	Kit mass under 5 kg (11 lb); carried hands-free while climbing	Weighing and climbing trial	Met on paper: 4.55 kg carried with the aluminium frame and the rope bag hauled up on the tag line (18C); whole kit 8.3 kg
R8	No damage to the palm	No bark cuts deeper than 5 mm after a full test lowering	Inspection after trials	Met on paper; to confirm by test (pads 1.9 MPa, sleeve 0.6 MPa at 2.5 kN)
R9	Cost	Parts under USD 850 per kit at single-kit prices, keeping the auto-locking descender and the evacuation triangle; group purchase by the training partner sought to lower it, one kit per climber group. Restated by Amish (19A); was USD 100	Costed bill of materials	Met on paper: USD 826 per kit (USD 24 under)
R10	Teachable	Climbers new to the kit complete a supervised rescue drill after a half-day session	Training trial with partner programme	Cannot be shown on paper; training trial at TRL 4

■ Assumptions

- A second climber is often on site or nearby when a climber is stranded, as in the reported cases.
- The rescuer can reach and work just below the victim using their own climbing device.

- Coconut bark and fibre can carry a collar load without crushing, if the load is spread.
- Victims can be fitted with a sling while hanging head down.

■ Safety

Safety: These requirements describe rope rescue equipment that holds a person at height. Meeting them on paper does not make the kit safe to use. Each is verified by test at TRL 4, and the kit is not used on a person until R1 has been proof-tested on cut trunk sections and the drill has been run with a dummy at low height.